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Refactor

Code Issues

Debugger

CODE

ANALYZE

Section Break

Run and Advance

Run to End

SECTION

Run

Step

St

RUN

simulation tas.m ×

untitled13.m ×

untitled14.m ×

untitled15.m ×

untitled16.m ×

+

/MATLAB Drive/untitled16.m

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clc;

clear;

close all;

% =====

% OBJECTIVE 1 : MIMO BER Performance

% =====

% SNR Range

SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO Parameters

Nt = 2;

Nr = 2;

BER = zeros(1,length(SNR));

for i = 1:length(SNR)

bits = randi([0 1],10000,1);

symbols = 2*bits - 1; % BPSK

H = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);

noise = (randn(Nr,10000)+1j*randn(Nr,10000))/sqrt(2+SNR(i));

tx = H(1,1)*symbols' + noise(1,:);

rx = real(tx)>0;

BER(i) = sum(rx'~=bits)/length(bits);

